CIS437 - Cloud Computing

– Cloud Networking Fundamentals –

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Based on the original version by Professor Erik Fredericks

Last Class

- Cloud service models
- Infrastructure as a Service (IaaS)
- Platform as a Service (PaaS)
- Serverless / Functions as a Service
- Software as a Service (SaaS)
- Control vs. management responsibility
- Choosing the right service model

Agenda

- Introduction to Networking in the Cloud
- VPCs and Subnets
- IP Addresses and CIDR
- Google Cloud Network Overview
- Routes and Firewall Rules
- Network Tags and Firewall Layers

Why Does a Cloud Application Need Networking?



Why Does a Cloud Application Need Networking?

Users → Load Balancer → Managed Instance Group → VMs

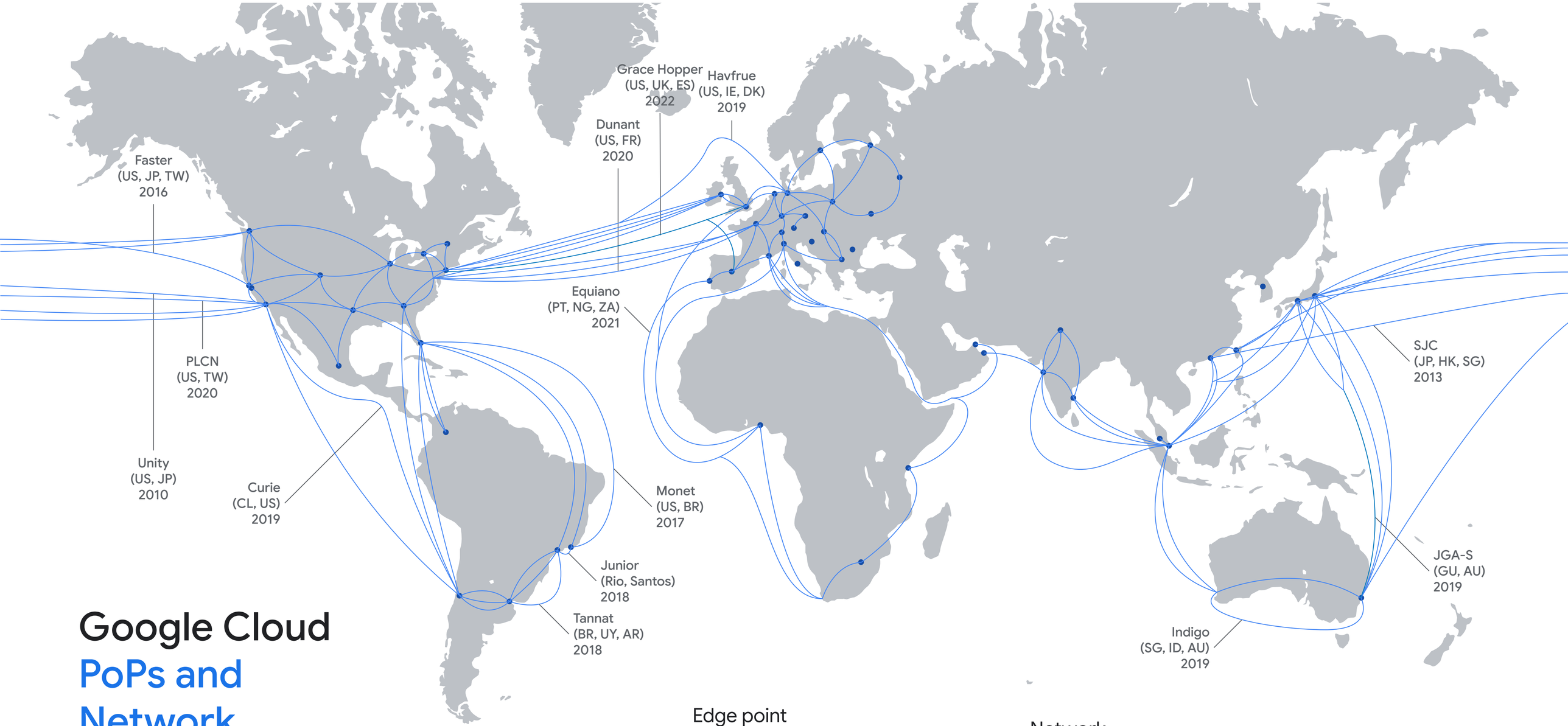
- Users need to reach cloud applications.
- Cloud resources need to communicate with each other.
- Networking controls **how traffic moves** and **what traffic is allowed**.

Cloud networking connects users, applications, and cloud resources.

A Background to Networking

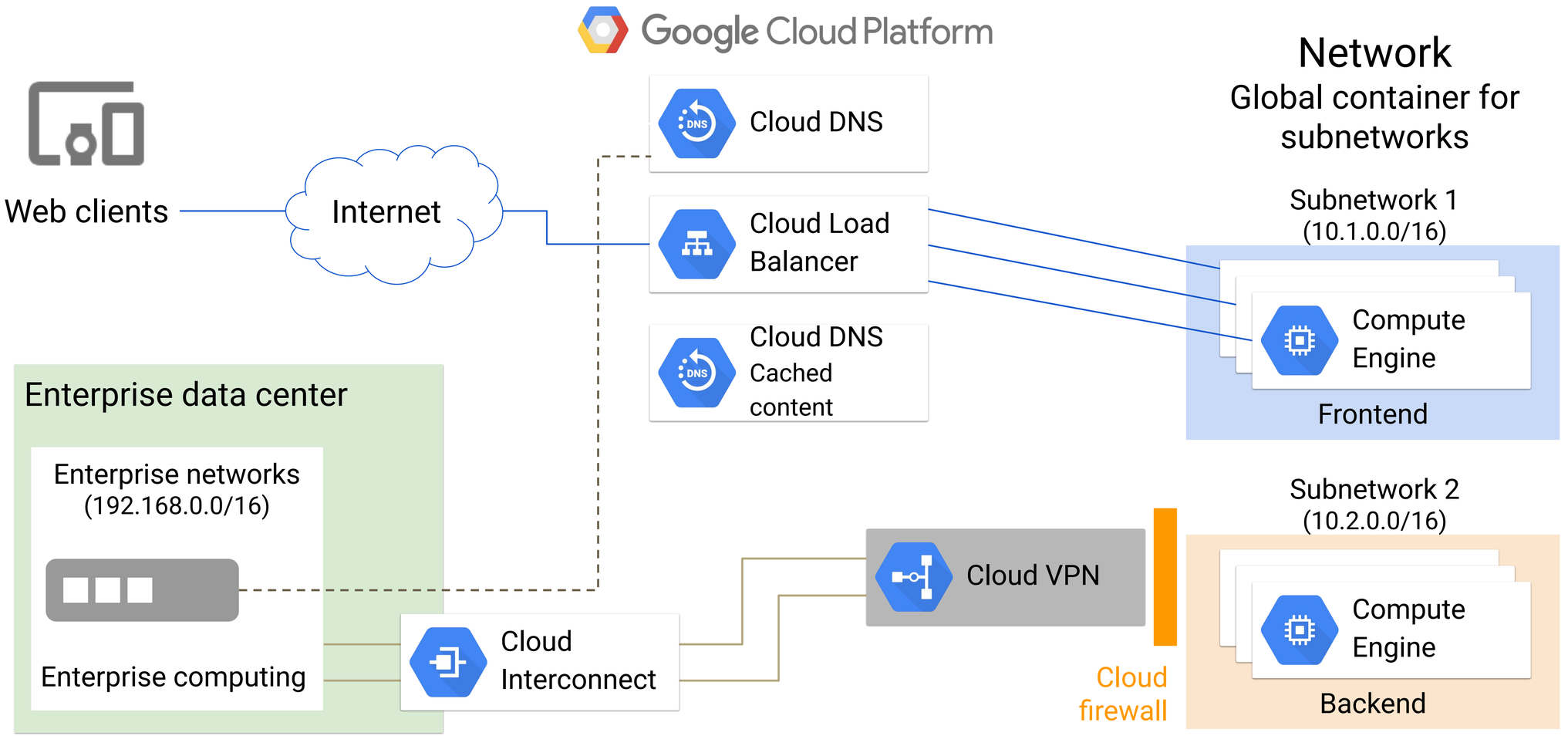


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Example: How Cloud Networking Components Work Together

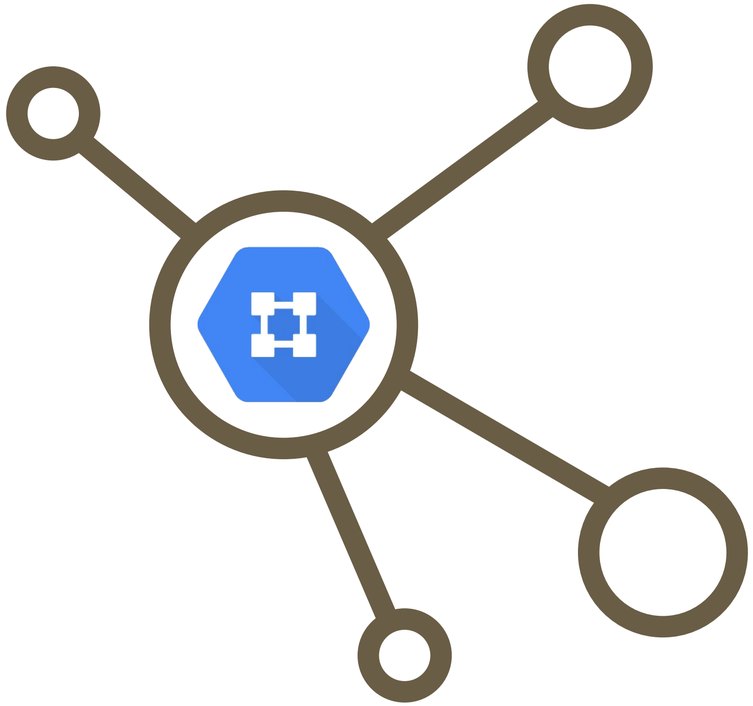


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What Is a VPC?

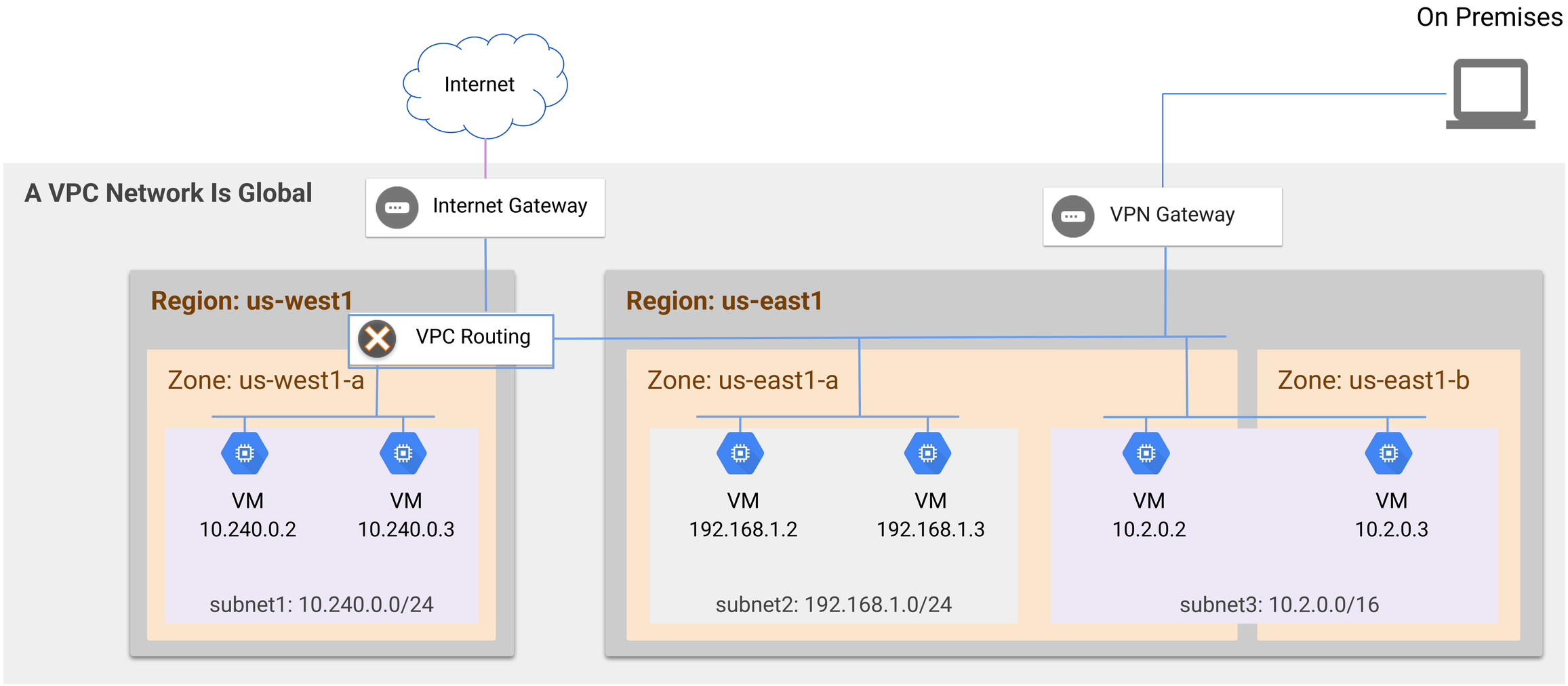
VPCs are software defined network (SDN) constructs

- Allow the deployment of IaaS resources
- No IP address ranges
- Global
- Contain subnets



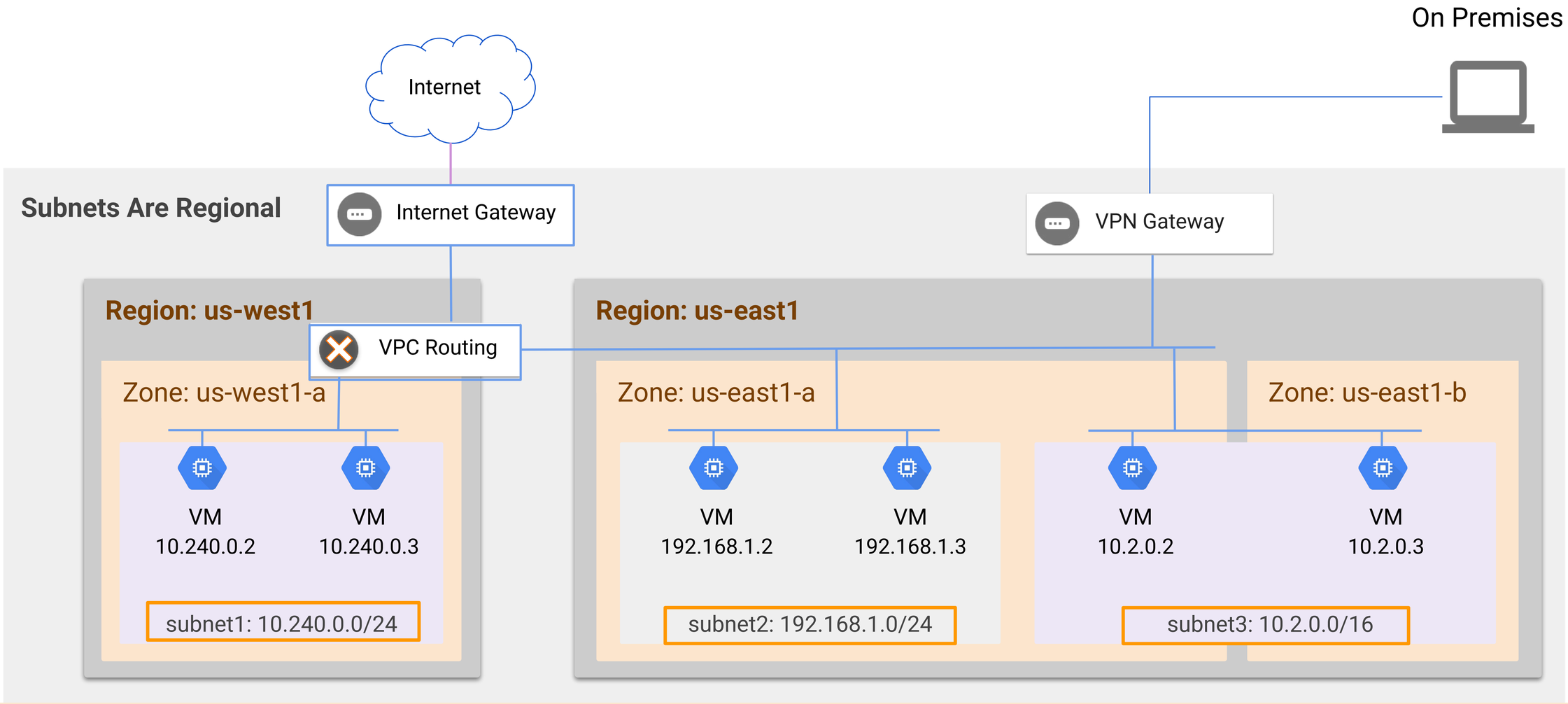
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A VPC network is a virtual version of a physical network and is a global resource



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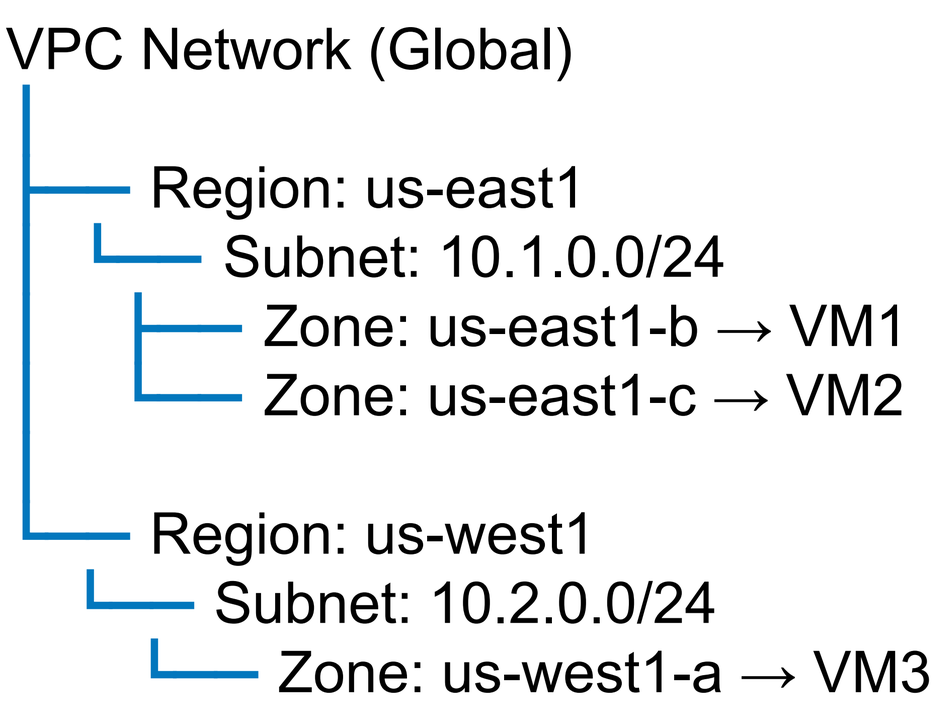
Subnets are regional and extend across zones in the same region



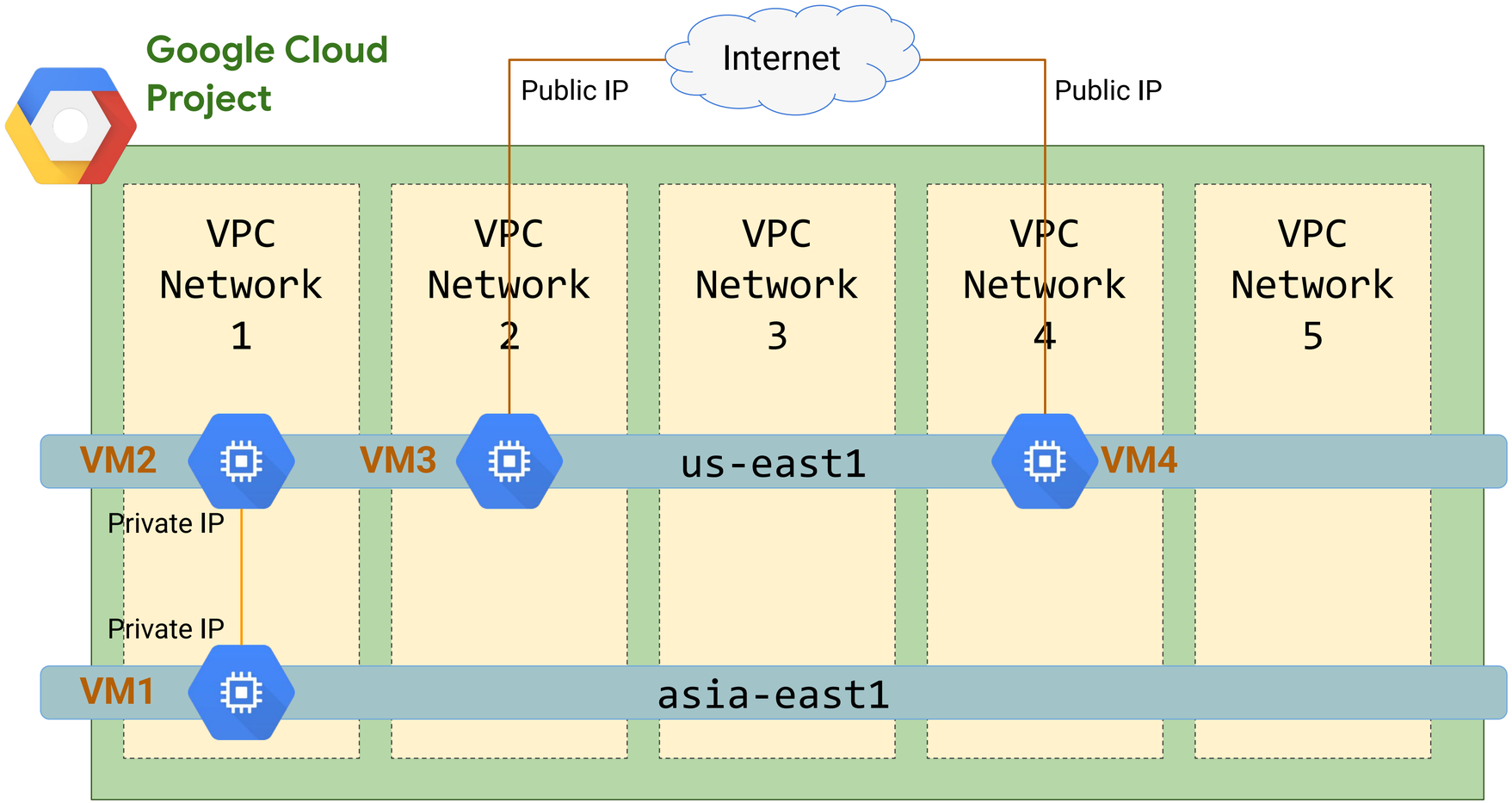
`10.240.0.0/24` roughly means all addresses beginning with `10.240.0.`

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VPC (global) → Subnet (regional) → VM (zonal)



Network behavior within a project



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Auto Mode vs. Custom Mode VPC Networks

Auto subnet mode

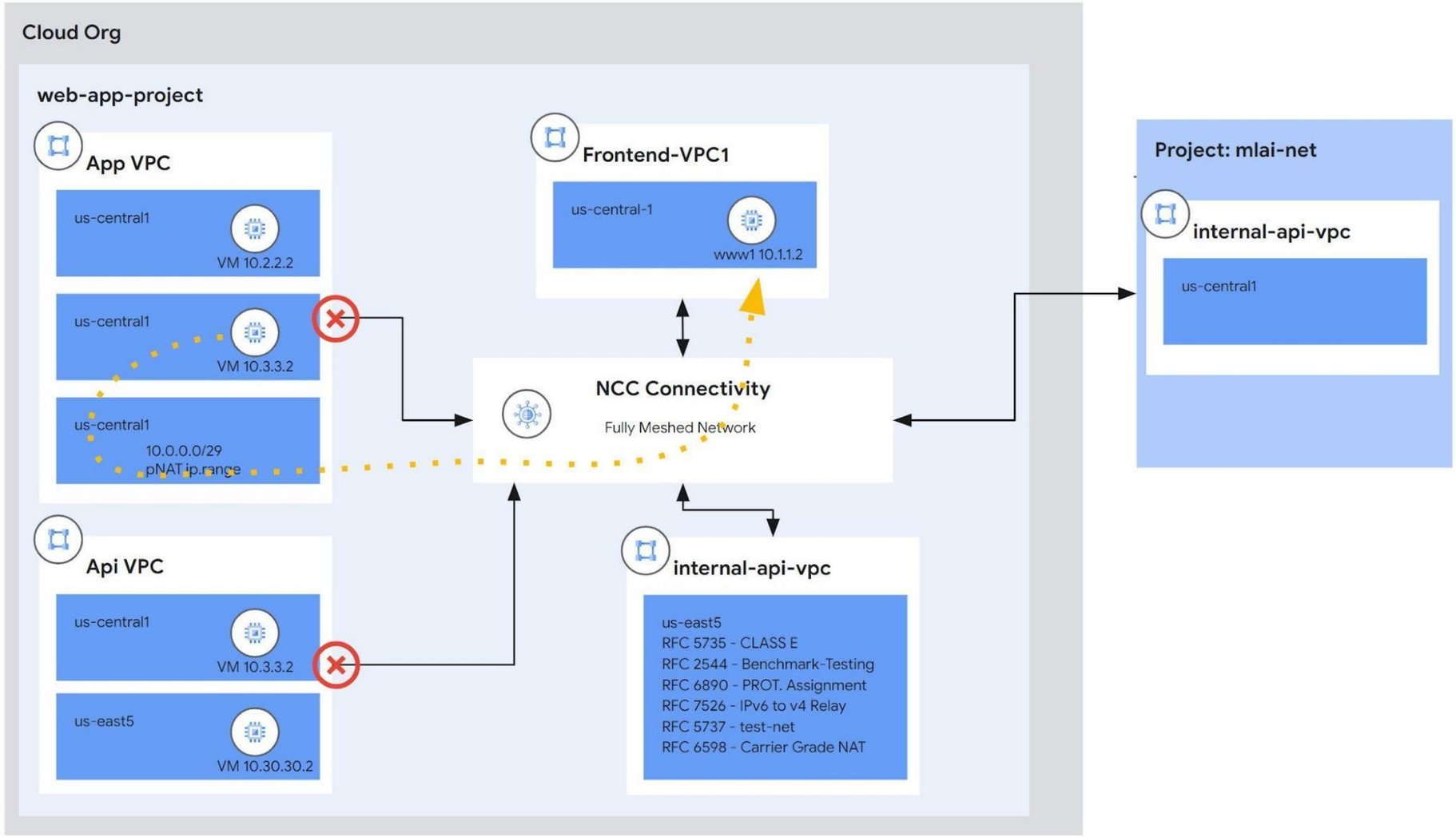
- One subnet from each region is automatically created
- Set of predefined IP ranges
- Comes with default firewall rules
- Expandable up to /16 only
- Good for isolated use cases (Proof of concepts (PoCs), testing, etc.)

Custom subnet mode

- No subnets are automatically created
- Subnets and IP ranges are defined
- No default firewalls rules
- Expandable to any RFC 1918 size
- Recommended for Production environments

Auto mode = easier setup | Custom mode = more control
Note: An auto mode VPC can be converted to custom mode, but the change is **one-way**.

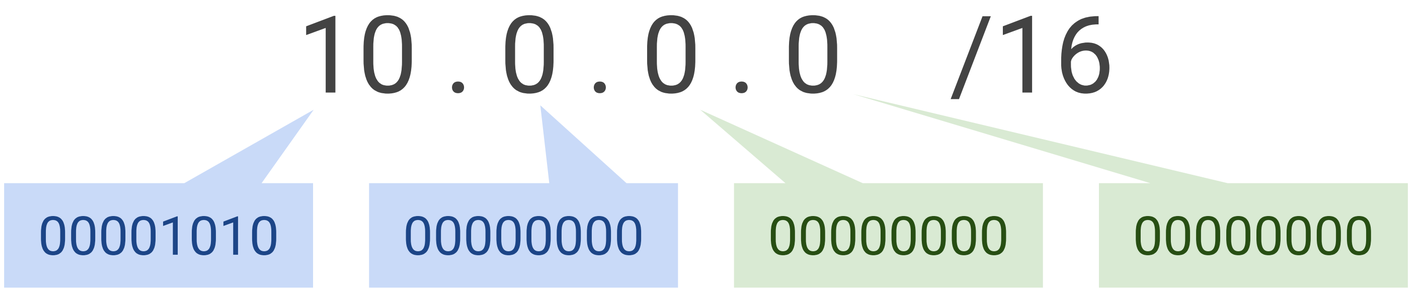
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Official Demos: Advanced Networking Demos

A VPC is made up of subnets

- Each subnet has an IP address range, written in CIDR notation
- CIDR = Classless Inter-Domain Routing
- An IPv4 address has 32 bits: four octets × 8 bits
- /16 means the first 16 bits are the network prefix
- The remaining bits identify addresses within that subnet



`10.0.0.0/16` → first two octets (`10.0`) are fixed

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A /16 range will provide 65,536 available IP addresses

- Total IPv4 addresses in the CIDR block
- Adding 1 to the /## will cut the available IP addresses in half.

RFC 1918 private IPv4 ranges

- 10.0.0.0/8
- 172.16.0.0/12
- 192.168.0.0/16

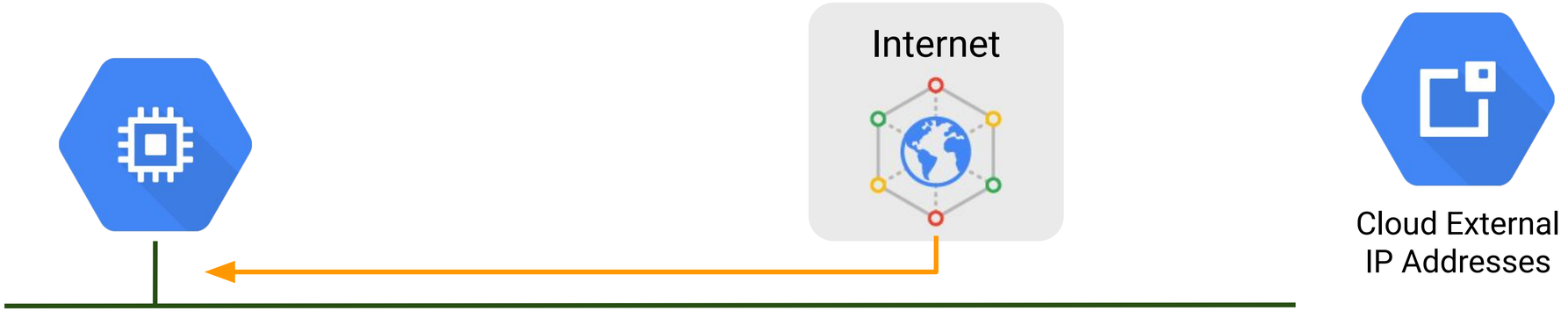
CIDR IP address totals

/16	/17	/18	/19	/20	/21	/22
65,536	**32,768**	**16,384**	**8,192**	**4,096**	**2,048**	**1,024**
/23	/24	/25	/26	/27	/28	
512	**256**	**128**	**64**	**32**	**16**	

Google Cloud note: In a primary subnet range, 4 addresses are reserved by Google Cloud.

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Public and Private IP address basics



Internal IP

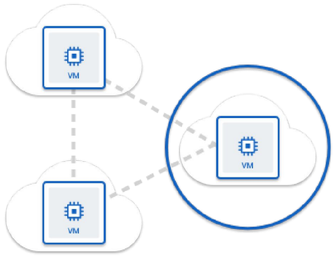
- Allocated from subnet range to VMs by DHCP.
- DHCP lease is renewed every 24 hours.
- VM name and IP is registered with network-scoped DNS.

External IP

- Can be assigned from pool (ephemeral) or reserved (static).
- Static external IP addresses may incur charges.
- VM doesn't know the external IP; it’s mapped to the internal IP.

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Key Google Cloud Networking Services



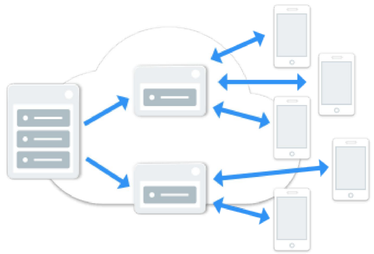
Virtual Private Cloud

Manage networking for resources



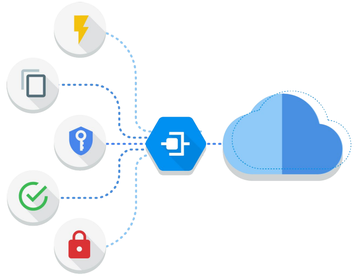
Cloud Load Balancer

Worldwide autoscaling and load balancing



Cloud CDN

Content delivery network



Cloud Interconnect

Fast, high availability interconnect



Cloud DNS

Highly available global DNS network

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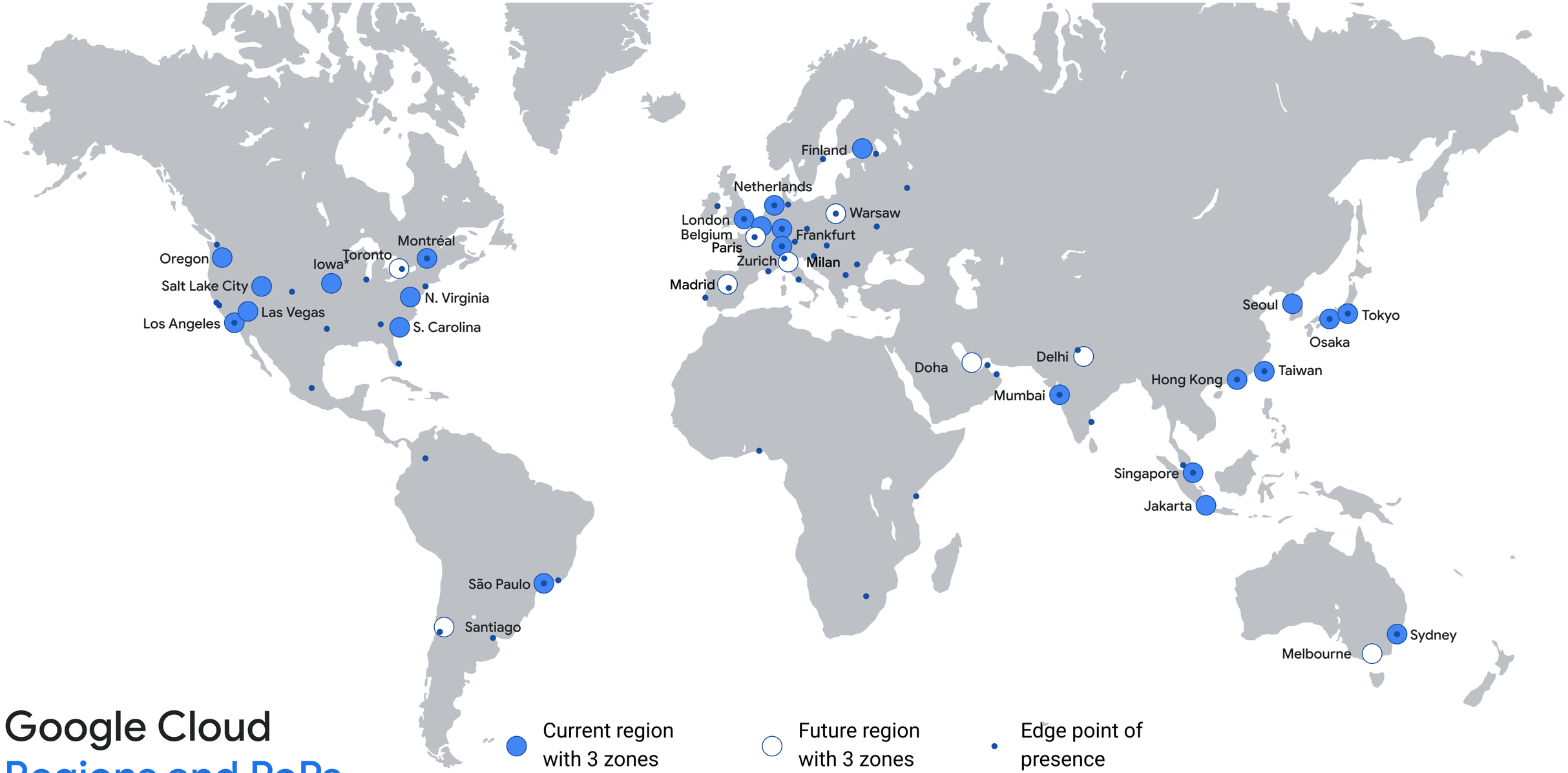
Google Cloud
Regions and Zones

Current region with 3 zones

Future region with 3 zones

* Exception: region has 4 zones.

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Google Cloud
Regions and PoPs

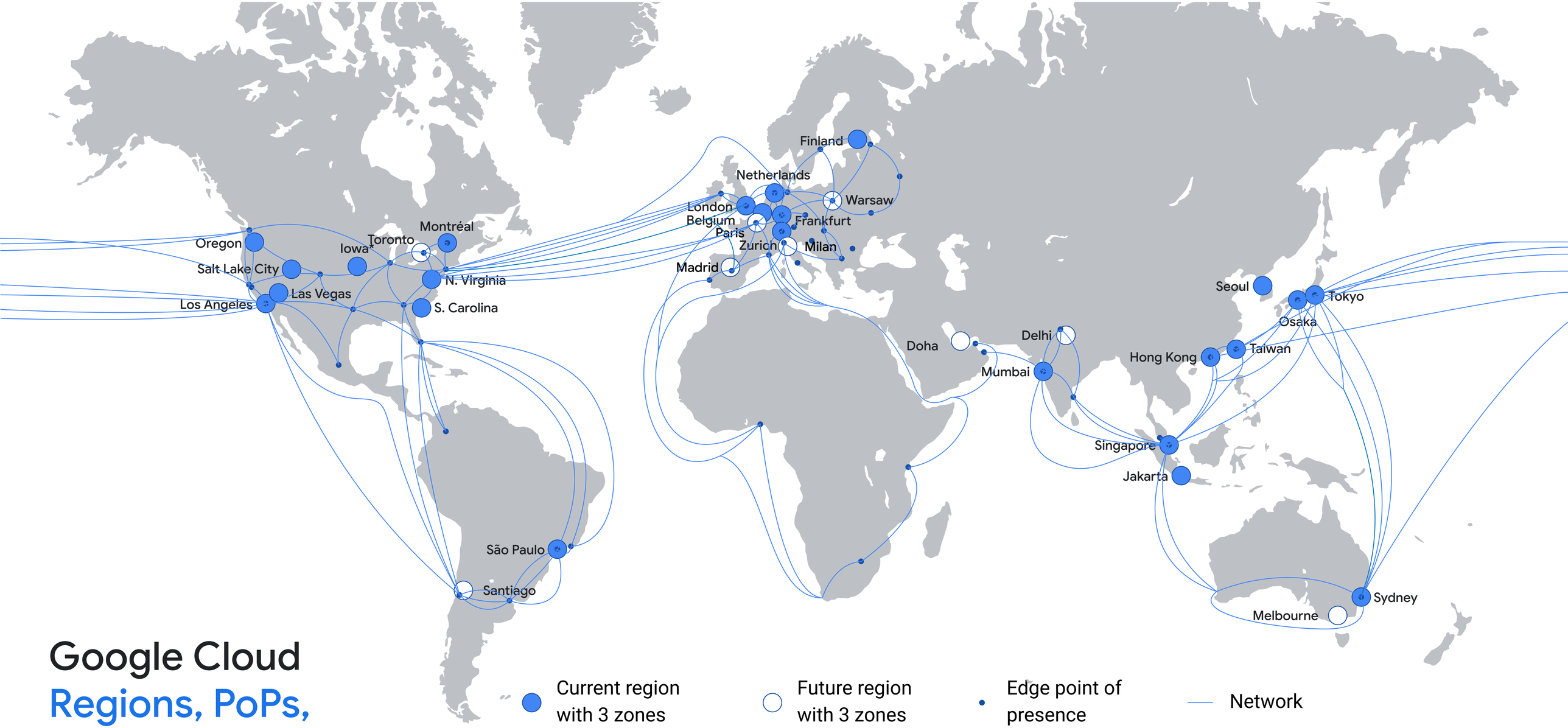
Current region with 3 zones

Future region with 3 zones

Edge point of presence

* Exception: region has 4 zones.

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Google Cloud
Regions, PoPs, and network

Current region with 3 zones

Future region with 3 zones

Edge point of presence

Network

* Exception: region has 4 zones.

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A route maps a destination IP range to a path / next hop

- Every network has routes that let instances in a network send traffic directly to each other.
- Every network has a default route that directs packets to destinations that are outside the network.
- Firewall rules must also allow the packet.

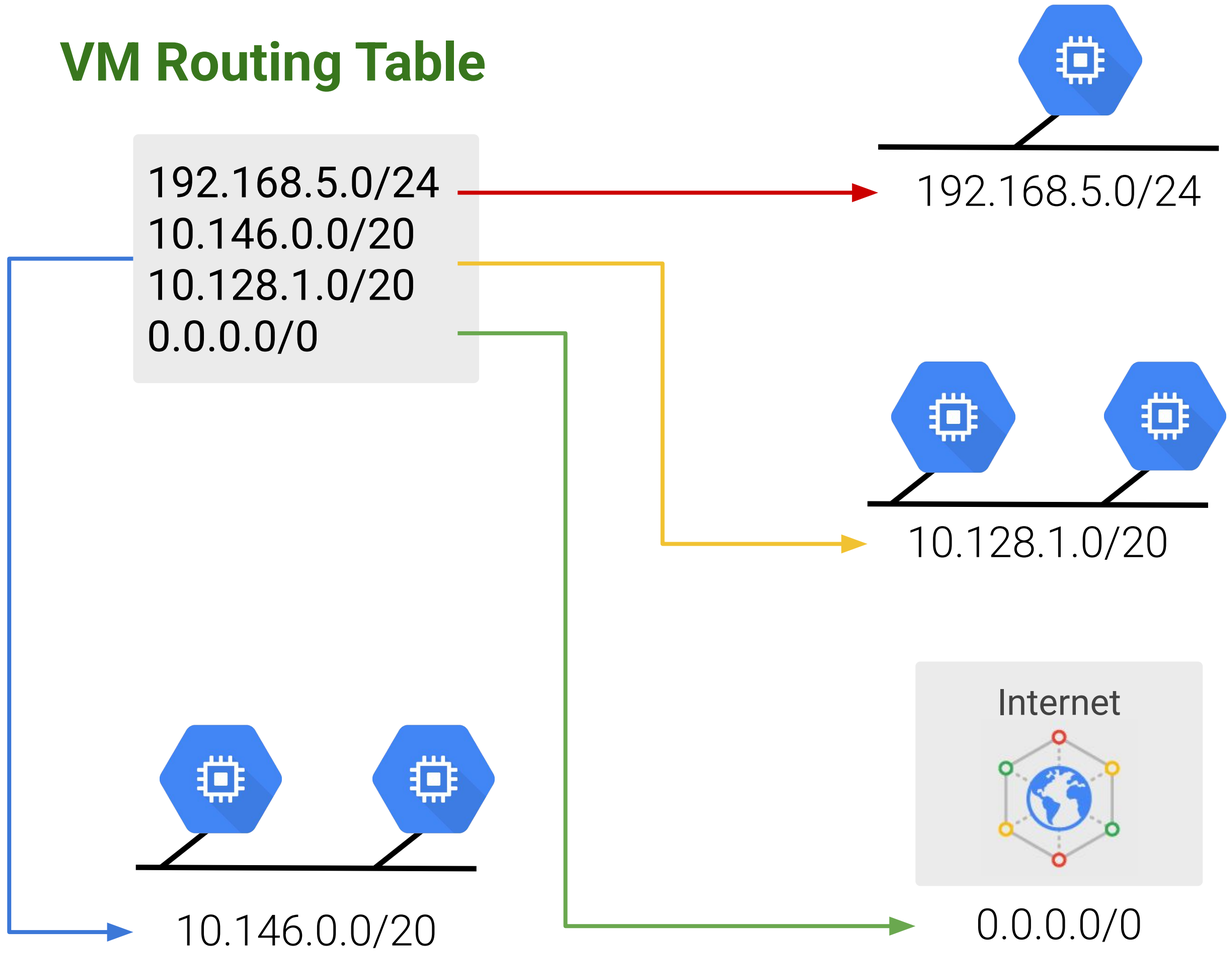


Route = Where should the traffic go?
Firewall = Is the traffic allowed?

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Routes Map Traffic to Destination Networks

Routing decision = look at the packet's destination IP address



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What Does 0.0.0.0/0 Mean?

```
/24 → 24 bits fixed

/16 → 16 bits fixed

/0  →  0 bits fixed
```

0.0.0.0/0 matches all IPv4 addresses

In a route

0.0.0.0/0 → default route for destinations not matched by a more specific route

In a firewall rule

Source 0.0.0.0/0 → traffic can originate from any IPv4 address

Firewalls protect virtual machine instances from unapproved connections

- A VPC network provides distributed firewall protection
- **Ingress** = incoming traffic
- **Egress** = outgoing traffic
- Rules are applied through the VPC, but traffic is allowed or denied for VM instances
- Firewall rules are **stateful**
- Implied rules:
  - Deny all ingress
  - Allow all egress



Route = Where should the traffic go?
Firewall = Is the traffic allowed?

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Express your desired firewall configuration as a set of firewall rules

Direction — ingress or egress

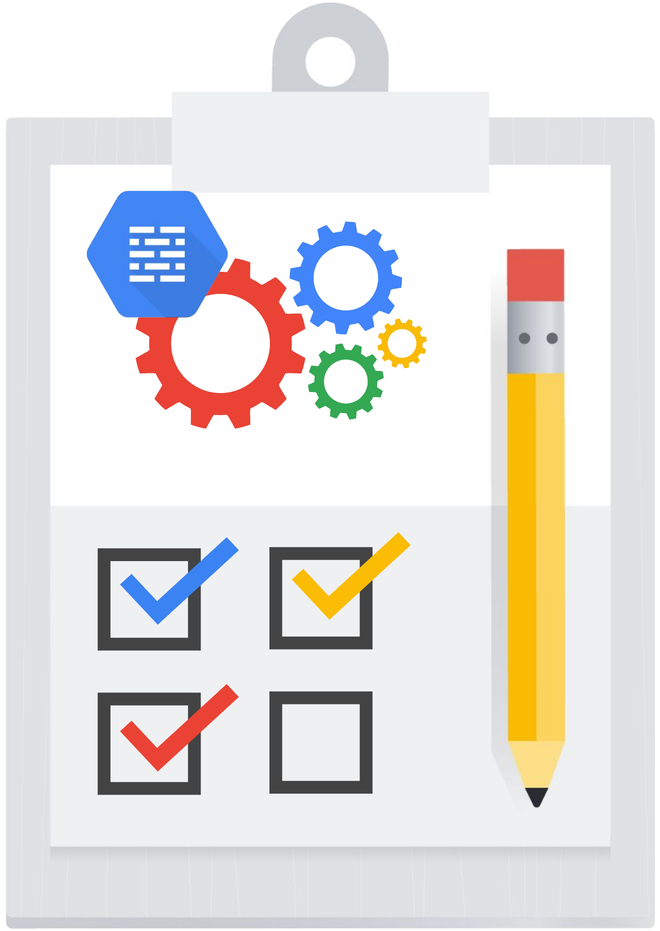
Source / Destination — where traffic comes from or goes to

Protocol and Port — for example TCP port 80

Action — allow or deny

Priority — lower number = higher priority

Target — which VM instances the rule applies to



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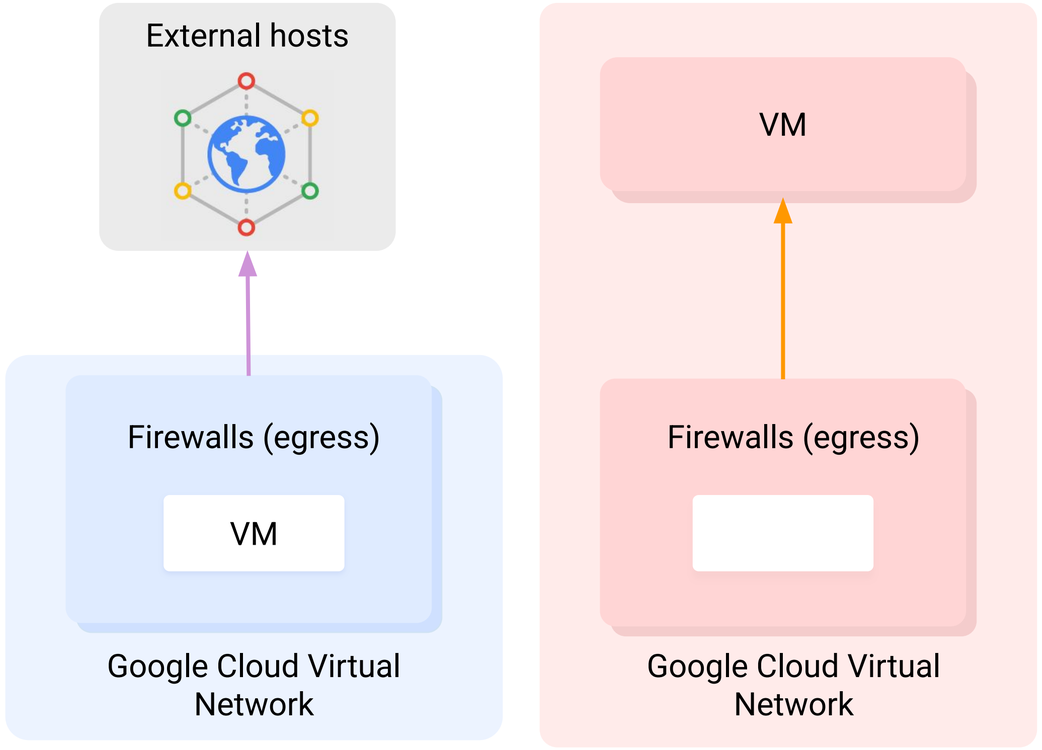
Egress Firewall Rules — Controlling Outgoing Traffic

Conditions:

- Destination CIDR ranges
- Protocols
- Ports

Action:

- **Allow:** permit the matching egress connection
- **Deny:** block the matching egress connection



Egress: VM → destination

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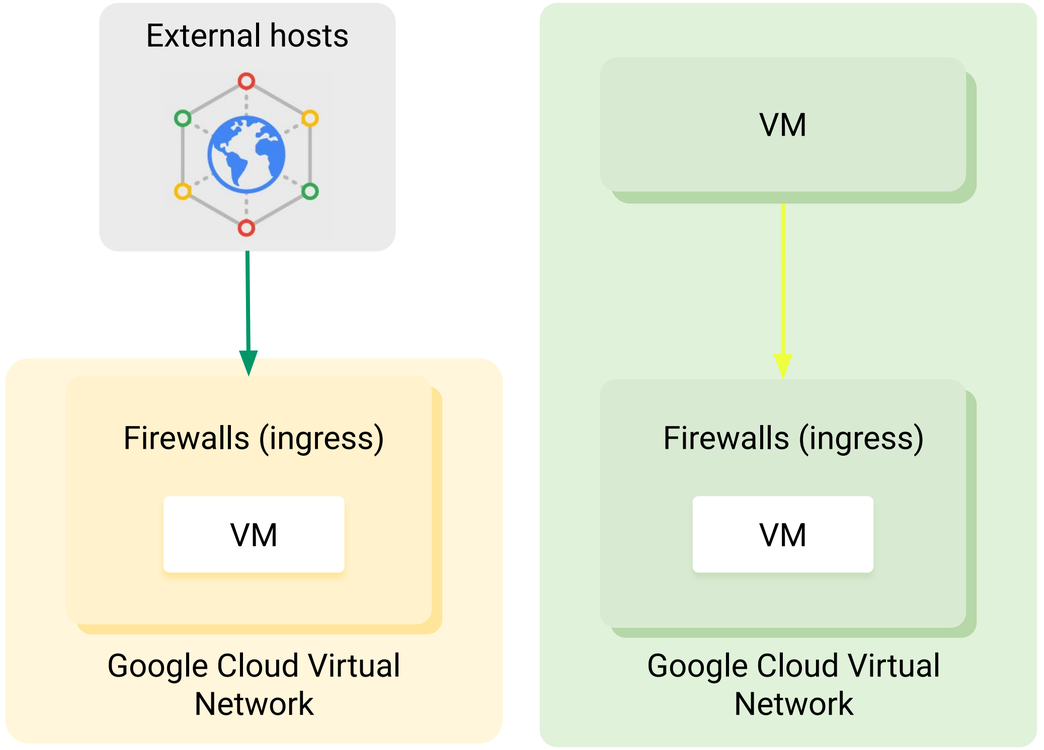
Ingress Firewall Rules — Controlling Incoming Traffic

Conditions:

- Source CIDR ranges
- Protocols
- Ports

Action:

- **Allow:** permit the matching ingress connection
- **Deny:** block the matching ingress connection



Ingress: source → VM

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Common Ports in Firewall Rules

Service	Protocol	Port	Typical use
SSH	TCP	**22**	Remote login to a VM
HTTP	TCP	**80**	Unencrypted web traffic
HTTPS	TCP	**443**	Encrypted web traffic

A firewall rule can allow or deny traffic based on protocol and port.

Summary

- A **VPC** is a global virtual network in Google Cloud.
- **Subnets are regional** and define IP address ranges using CIDR.
- **Internal IPs** support private communication; **external IPs** support Internet-facing communication.
- **Routes** determine where network traffic goes.
- 0.0.0.0/0 represents **all IPv4 addresses**.
- **Firewall rules** determine whether traffic is allowed or denied.
- **Ingress** controls incoming traffic; **egress** controls outgoing traffic.
- Firewall rules can use **protocols and ports**, such as TCP 22, 80, and 443.

Practice: Configure a Web Server

Scenario

You have a VPC with a subnet in us-east1:

10.10.0.0/24

A VM named **web-server** has:

- Internal IP: 10.10.0.5
- External IP: assigned
- Apache running on TCP port **80**
- SSH running on TCP port **22**

- Is the **VPC**, **subnet**, or **VM** global/regional/zonal?
- How many total IPv4 addresses are in 10.10.0.0/24?
- What ingress firewall rule should we create for **HTTP**?
- What ingress firewall rule should we create for **SSH**?
- If Apache is running but TCP port 80 is blocked by the firewall, can users access the website?



Requirements:

1. Anyone on the Internet should be able to access the website.
2. SSH should only be allowed from the administrator's IP: 203.0.113.10.